

GenAI FastAPI Backend

This project is a backend application built with FastAPI, providing a provider-agnostic LLM gateway and a multi-source i

SEVERITY 46	CONFIDENCE 70	HEALTH 0
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Language	Python	Architecture	Monolith with modular components
Complexity	medium	Sec Posture	—
Bugs Found	7	Vulns Found	0
Doc Grade	C	Prod Ready	—

No summary available.

Project	GenAI FastAPI Backend
Purpose	This project is a backend application built with FastAPI, providing a provider-agnostic LLM gateway and
Architecture	Monolith with modular components
Languages	Python, Markdown, JSON, Text
Frameworks	FastAPI
Complexity	medium
Entry Points	app/main.py
Notes	The project exhibits a modular monolithic architecture, with a clear separation of concerns between diff

Code Quality: **fair** · Testing Gaps: The code lacks unit tests for edge cases and scenarios, and does not cover all p

Uncaught Exception in genai-fastapi-backend/app/api/routes/chat.py ■ genai-fastapi-backend/app/api/routes/chat.py Category: unhandled_exception The code does not handle the exception properly, it just raises it again without any additional information. <pre>raise</pre> ■ Fix: Handle the exception properly by logging the error and returning a meaningful error message to the user.	HIGH
Potential Null Reference in genai-fastapi-backend/app/agents/session16/orchestrator.py ■ genai-fastapi-backend/app/agents/session16/orchestrator.py Category: null_reference The code does not check if the 'validation' object is null before accessing its properties. <pre>if not validation.approved and result.output.get("skipped")</pre> ■ Fix: Add a null check for the 'validation' object before accessing its properties.	MEDIUM
Logic Error in genai-fastapi-backend/app/agents/session15/executor.py ■ genai-fastapi-backend/app/agents/session15/executor.py Category: logic_error The code does not handle the case where the 'output' dictionary does not contain an 'error' key. <pre>error=output.get("error") if not success else None</pre> ■ Fix: Add a check to handle the case where the 'output' dictionary does not contain an 'error' key.	MEDIUM
Potential Type Error in genai-fastapi-backend/data/session11/eval_questions.json ■ genai-fastapi-backend/data/session11/eval_questions.json Category: type_error The code does not specify the type of the 'question' and 'ground_truth' fields in the JSON object. <pre>"question": "What are the interview rounds at Amazon for an SDE role?"</pre> ■ Fix: Specify the type of the 'question' and 'ground_truth' fields in the JSON object.	LOW
Potential Race Condition in genai-fastapi-backend/app/agents/session17/playground.py ■ genai-fastapi-backend/app/agents/session17/playground.py Category: race_condition	HIGH

The code uses a global variable 'flaky_count' which can be accessed by multiple threads.

```
global flaky_count
```

■ **Fix:** Use a thread-safe way to access and modify the 'flaky_count' variable.

Potential Anti-Pattern in genai-fastapi-backend/app/agents/session15/validator.py

LOW

■ genai-fastapi-backend/app/agents/session15/validator.py

Category: anti_pattern

The code uses a complex conditional statement which can be hard to read and maintain.

```
if not result.success: ...
```

■ **Fix:** Simplify the conditional statement to make it easier to read and maintain.

Potential Unhandled Exception in genai-fastapi-backend/app/agents/session17/policy.py

MEDIUM

■ genai-fastapi-backend/app/agents/session17/policy.py

Category: unhandled_exception

The code does not handle the case where the 'evaluation' object is null.

```
if evaluation.score == 0.0:
```

■ **Fix:** Add a null check for the 'evaluation' object before accessing its properties.

Critical	High	Medium	Low
0	0	0	0

✓ No vulnerabilities detected.

Overall Grade	C
README Score	60/100
Docstring Coverage	medium

Quick Wins

- Add docstrings to all functions and classes
- Create a contributing guide with code style and commit message guidelines
- Add a license file to the project

README Improvements

Priority	Section	Suggestion
HIGH	Introduction	Add a brief overview of the project, its purpose, and its goals.
HIGH	Getting Started	Create a step-by-step installation guide for the project.
MEDIUM	API Documentation	Add API documentation for all endpoints, including usage examples and response formats.

This report was generated automatically by the AI Code Review Platform using Groq Llama 3.3, LangGraph multi-agent orchestration, and ChromaDB RAG retrieval. All findings should be reviewed by a qualified engineer before remediation.